

8.10.2026

PRACTICAL - 12

END-END COMMUNICATION AT

TRANSPORT LAYER

AIM:-

To implement echo client server using TCP/UDP sockets.

TCP:-

CODE:-

import socket

import threading

def start_server():

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(('127.0.0.1', 5000))

server.listen(1)

print("[SERVER] Listening on 127.0.0.1:5000")

conn, addr = server.accept()

print(f"[SERVER] connected by {addr}")

while True:

data = conn.recv(1024).decode()

if not data or data.lower() == 'quit':

print("[SERVER] Connection closed by client.")

break

print(f"[SERVER] Received: {data}")

conn.send(f"Server received: {data}".encode())

conn.close()

server.close()

def start_client():

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(('127.0.0.1', 5000))

print("[CLIENT] Connected to server")

while True:

msg = input("Enter message (or 'quit' to exit): ")

client.send(msg.encode())

if msg.lower() == 'quit':

break

response = client.recv(1024).decode()

print(f"[SERVER RESPONSE] {response}")

client.close()

```

if __name__ == "__main__":
    server_thread = threading.Thread(
        target=start_server,
        daemon=True
    )
    server_thread.start()
    start_client()

```

OUTPUT:-

```

python practical12-udp.py
[SERVER] Listening on 127.0.0.1:5000
[CLIENT] Connected to server
[SERVER] Connected by ("127.0.0.1", 5050)
Enter message (or 'quit' to exit): Computer Networks
[SERVER] Received: Computer Networks
[SERVER RESPONSE] Server received: Computer Networks
Enter message (or 'quit' to exit): quit
[SERVER] connection closed by client.

```

UDP:

Code:-

```

import socket
import threading

def udp_server():
    s = socket.socket(
        socket.AF_INET,
        socket.SOCK_DGRAM
    )
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")

```

```
def udp-client():
```

```
    c = socket.socket(  
        socket.AF_INET,  
        socket.SOCK_DGRAM  
    )
```

```
    server = ('127.0.0.1', 12000)
```

```
    while True:
```

```
        msg = input('Enter message (or 'quit' to stop): "')
```

```
        c.sendto(msg.encode(), server)
```

```
        if msg.lower() == 'quit':
```

```
            print("[CLIENT] closing connection...")
```

```
            break
```

```
        data, _ = c.recvfrom(1024)
```

```
        print("Echo from server: ", data.decode())
```

```
    c.close()
```

```
    print("[CLIENT] stopped.")
```

```
if __name__ == "__main__":
```

```
    threading.Thread(  
        target = udp_server,  
        daemon = True  
    ).start()
```

```
    udp-client()
```

OUTPUT:-

python practical12-udp.py

Enter message (or 'quit' to stop): [SERVER] Running on 127.0.0.1:12000

Computer Networks

[SERVER] received from ('127.0.0.1', 55641): Computer Networks

Echo from server: Computer Networks

Enter message (or 'quit' to stop): quit

[CLIENT] closing connection...

[SERVER] client('127.0.0.1', 55641) ended the connection.

[CLIENT] stopped.

[SERVER] stopped.

RESULT:-

Thus end-to-end communication is implemented successfully at transport layer.